Brain Tumor Detection using AI

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Abstract:

Cancer is one of the leading problems in the world today. There are about 18.1 million cancer patients in the world today. Cancer can be difficult to detect in the early stages. We can use Artifical Intelligence to find out patterns in the reports of patients in the early stages and help in the prevention of cancerous tumors from growing any further. Artificial intelligence is a computer program or algorithm that uses past data to make future predictions. It teaches itself how to analyze and interpret data, which helps machine learning algorithms may pick up on patterns that are not readily discernable to the human eye or brain. As these algorithms are exposed to more new data, their ability to learn and interpret the data improves. Researchers have also used deep learning, a type of machine learning, in cancer imaging applications. Deep learning refers to algorithms that classify information in ways much to the human brain does. Deep learning tools use "artificial neural networks" that mimic how our brain cells take in, process, and react to signals from the rest of our body. Research on AI for cancer imaging: Studies suggest that AI has the potential to improve the speed, accuracy, and reliability with which doctors answer those questions. But what scientists are most excited about is the potential for AI to go beyond what humans can currently do themselves. AI can "see" things that we humans can't, and can find complex patterns and relationships between very different kinds of data.

Items/Keywords:

- Artificial intelligence
- Cancerous tumor
- Machine Learning Algorithms
- Neural Networks
- Deep learning

Contents:

Cancer is one of the leading problems in the world today. There are about 18.1 million cancer patients in the world today. Cancer can be difficult to detect in the early stages. We can use Artificial Intelligence to find out patterns in the reports of patients in the early stages and help in the prevention of cancerous tumors from growing any further.

Artificial intelligence is a computer program or algorithm that uses past data to make future predictions. It teaches itself how to analyze and interpret data, which helps machine learning algorithms pick up on patterns that are not readily discernible to the human eye or brain. As these algorithms are exposed to more new data, their ability to learn and interpret the data improves. Researchers have also used deep learning, a type of machine learning, in cancer imaging applications. Deep learning refers to algorithms that classify information in ways much as the human brain does. Deep learning tools use "artificial neural networks" that mimic how our brain cells take in, process, and react to signals from the rest of our body. Research on AI for cancer imaging: Studies suggest that AI has the potential to improve the speed, accuracy, and reliability with which doctors answer those questions. But what scientists are most excited about is the potential for AI to go beyond what humans can currently do themselves. AI can “see” things that we humans can’t, and can find complex patterns and relationships between very different kinds of data.

AI can “see” things that we humans can’t, and can find complex patterns and relationships between very different kinds of data. AI can be more reliable when it comes to identifying if a tumor is cancerous or not. The accuracy of AI is reliable enough to be used in real life. This can help radiologists with the identification of tumors. This can also help the doctors, as AI can help to identify some specific features or symptoms of the tumor. This can further help doctors to learn about Cancer.

As a result, the radiology department has gained prominence in the study of brain tumors using imaging methods. Many studies have looked at the causes of brain tumors, but the results have not been conclusive. FCM helps to increase accuracy. Amato et al. structured PC-assisted recognition using mathematical morphological reconstruction (MMR) for the initial analysis of brain tumors. Test results show the high accuracy of the segmented images while significantly reducing the time of calculation. A classification of neural deep learning systems was proposed for identifying brain tumors. Discrete wavelet transformation (DWT), excellent extraction methods, and main component analysis (PCA) were applied to the classifier here, and performance evaluation was highly acceptable across all performance measurements. In , a new classifier system was developed for brain tumor detection. The proposed system achieved 92.31% of accuracy. In , a method was suggested for classifying the brain MRI images using an advanced machine learning approach and brain structure analytics. To identify the separated brain regions, this technique provides greater accuracy and finds the ROI of the affected area. Researchers in [proposed a strategy to recognize MR brain tumors using a hybrid approach incorporating DWT transform for feature extraction, a genetic algorithm to reduce the number of features, and to support the classification of brain tumors by vector machine (SVM). Specific segmentation concepts include region-based segmentation edge-based techniques and thresholding techniques for the detection of cancer cells from normal cells. The common classification method is based on the Neural Network Classifiers, SVM Classifier, and Decision Classifier. In, a brain tumor detection method was developed using the GMDSWS-MEC model. The result shows high accuracy and less time to detect tumors.

Cancer is the leading cause of death in the world. This disease is characterized by the development of abnormal cells that divide uncontrollably and have the ability to infiltrate and destroy normal body tissue. Cancer often has the ability to spread throughout your body. Signs and symptoms caused by cancer will vary depending on what part of the body is affected. Some general signs and symptoms associated with, but not specific to, cancer, include:
- Fatigue.
- Lump or area of thickening that can be felt under the skin.
- Weight changes, including unintended loss or gain.
- Changes in bowel or bladder habits

Cancer is caused by changes (mutations) to the DNA within cells. The DNA inside a cell is packaged into a large number of individual genes, each of which contains a set of instructions telling the cell what functions to perform, as well as how to grow and divide. Errors in the instructions can cause the cell to stop its normal function and may allow a cell to become cancerous. This is how cancer is formed in the body.

The technique used for making this model work is CNN (Convolutional Neural Network). The different stages by which CNN is applied to the dataset are:

1. First, the required packages are imported.
2. Then the folder where the dataset is stored is imported.
3. Image reading is done after that it is labeled (such as 0 for benign and 1 for malicious) and then the reports are stored in the data frame.
4. Then the size of the image is changed to 224*224 and the shape of the image is.
5. Image normalization is a complete training dataset and a validation dataset was used for evaluating the model.
6. Then the accuracy of the model is evaluated using the test images.
7. The loss graph and the accuracy graph are plotted. The entire implementation is done using python 3.8 and is executed on Google Colaboratory and the model is stored in Google Drive

Results:

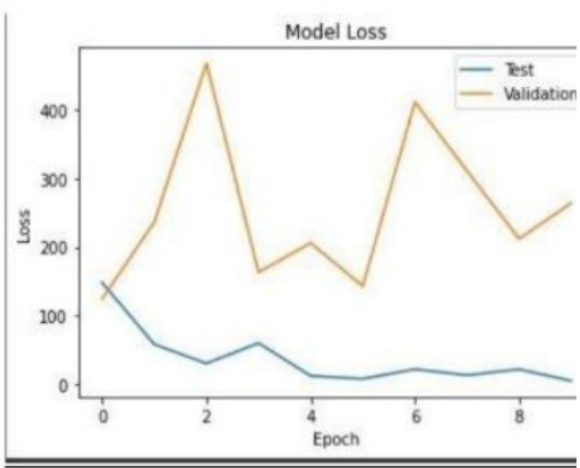


Figure 3. Model loss

When the model is applied to the testing data set for 10 epochs, a validation accuracy of 82.86% is obtained and the validation loss is also less. As seen in figure 3, when the model is applied to the validation, then a high loss is obtained but once applied to the testing set, the loss gradually decreases with the increasing number of epochs.

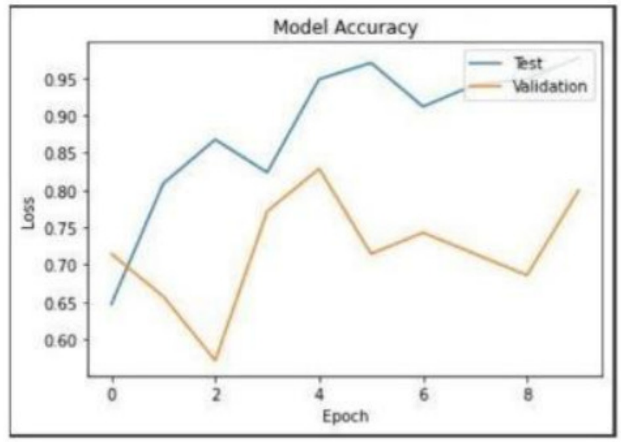
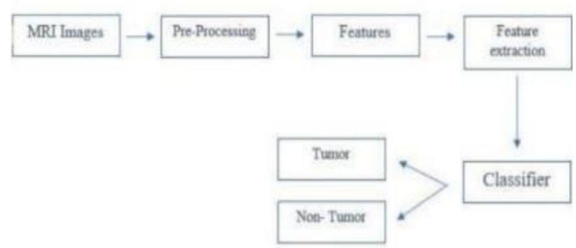


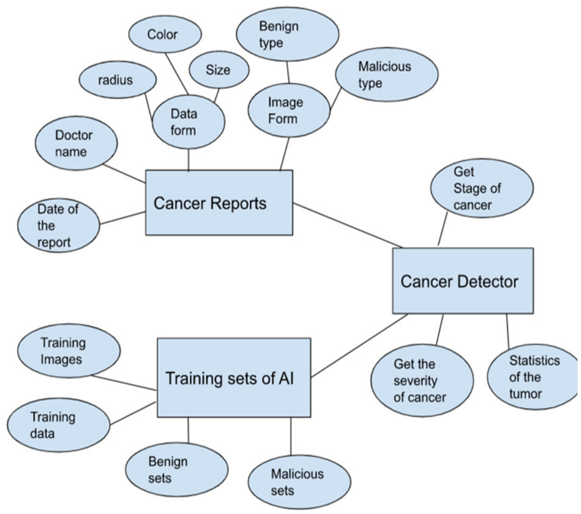
Figure 4. Model accuracy

The accuracy of the convolutional neural network model achieved after applying it to the testing set was 97.79%. with a very minimal loss with increasing epochs. The difference in model accuracy can be seen between the validation dataset and the training dataset.
System architecture:

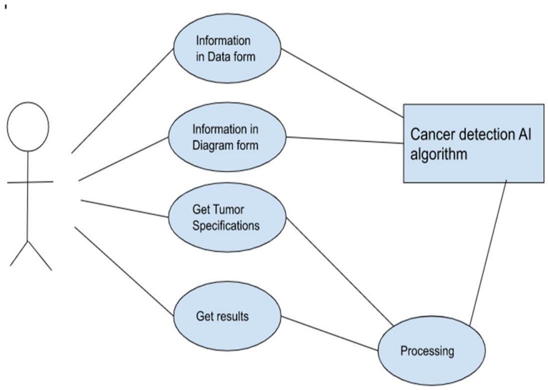


We collect the information either in the form of MRI images or MRI reports and then Pre-process them then the AI will detect some features or patterns in the data which is based on the data entered into the AI algorithm in the training sets. These Features will be classified by the classifier as either tumourous or non-tumourous.

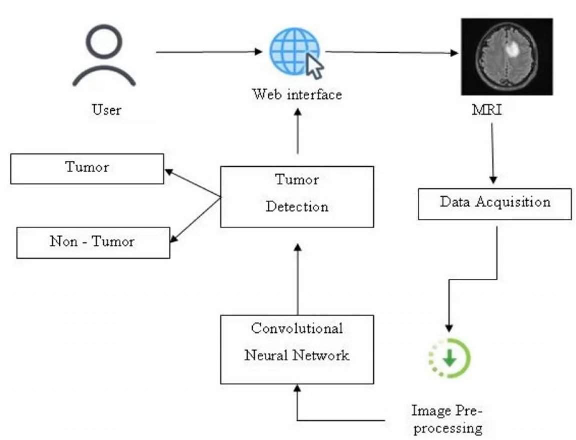
ER Diagram:



Use case Diagram

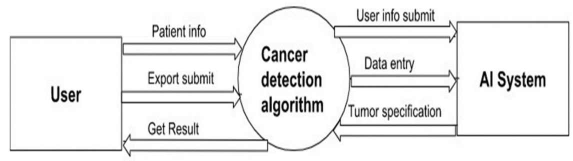


Activity Diagram

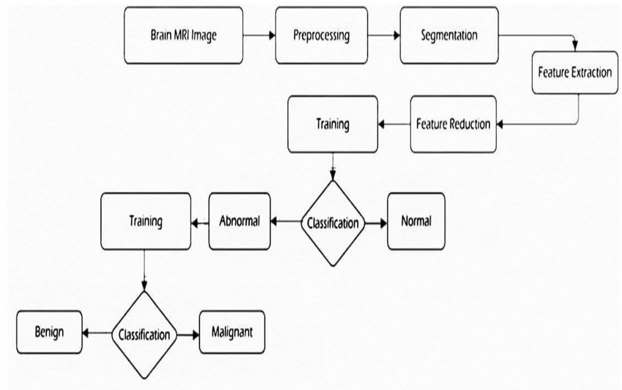


Data Flow Diagram:

Level 0



Level 1



Conclusion:

There are various machine learning algorithms that can be used to predict the state of a tumor based on MRI scans. One popular algorithm is the convolutional neural network (CNN), which is a type of deep learning algorithm.

To train a CNN for tumor classification, a dataset of labeled MRI scans is required. The dataset should include examples of both benign and malignant tumors, as well as images with various levels of complexity and quality. The CNN is then trained on the dataset, with the goal of minimizing the error between the predicted and actual tumor classification.

Overall, using a CNN algorithm to predict the state of a tumor from MRI scans can provide doctors with a reliable and accurate tool for diagnosis and treatment planning